

Sucrose activates human taste pathways differently from artificial sweetener.

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Animal models suggest that sucrose activates taste afferents differently than non-caloric sweeteners. Little information exists how artificial sweeteners engage central taste pathways in the human brain. We assessed sucrose and sucralose taste pleasantness across a concentration gradient in 12 healthy control women and applied 10% sucrose and matched sucralose during functional magnet resonance imaging. The results indicate that (1) both sucrose and sucralose activate functionally connected primary taste pathways; (2) taste pleasantness predicts left insula response; (3) sucrose elicits a stronger brain response in the anterior insula, frontal operculum, striatum and anterior cingulate, compared to sucralose; (4) only sucrose, but not sucralose, stimulation engages dopaminergic midbrain areas in relation to the behavioral pleasantness response. Thus, brain response distinguishes the caloric from the non-caloric sweetener, although the conscious mind could not. This could have important implications on how effective artificial sweeteners are in their ability to substitute sugar intake.